

Pertinent Thoughts — Scout (530 College Basketball Pipeline)

This document mirrors **Pertinent_Thoughts.md** (Army / OER work): important discoveries, reflections on code and results, open problems, and directions worth investigating for the **Scout** dissertation thread — ESPN box → SR advanced → player–season panel → leave-one-out teammate pool quality → draft outcome EDA.

How to use it: Add dated entries or new `##` sections as you go. Each block can follow the template: **Topic** → **Content to consider including** → **Potential placement** → **Key points** (plus optional implementation notes).

Generative Tier 1 Lab (538D, June 2026)

Topic: Soft assignment + congestion selection vs empirical 530 conditioning.

Key points:

- **Thread A implemented:** `tier1_pool_assignment.py`, CELL 10–12; $\tau \approx 0.65$; Plot A calibrated to 530 CELL 8 overlap.
 - **Minimal score:** $S_i = A_i - wL_C$ with `crowding_smooth`; z-scored ability needs `crowding_l_z_scale` (p90–p10 of A).
 - **539 vs 538D:** 539 is a **bundled DGP** (sort-chop, score noise, 90th percentile); 538D is a **modular lab** anchored to 530 — not parent/child decomposition.
 - **Plot B axis:** `SHOW_PLOT_B_TEAM_MEAN` — inverted-U on `team_mean`; mostly **decreasing** on L_Q LOO with same knobs.
 - **Status doc:** `5-Manuscript/Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md`.
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SR–ESPN School Slugs, Small Colleges, and Coverage Equity

Topic: Unmatched schools after `bpm_merge.run_match`; manual crosswalk / alias loop; re-scrape behavior.

Content to consider including:

Many unmatched player–season rows cluster on **small religious colleges, HBCUs, NAIA-style programs, and renamed schools** where the heuristic `school_slug` (from `team_short_display_name`) does not match Sports Reference’s URL segment. The pipeline already surfaces this in `bpm_panel_rows_unmatched.csv` and optional `print_unmatched_school_lists` (aggregated by `team_id` / slug). Fixing mappings lives in `DO_NOT_ERASE/sr_school_slug_crosswalk.csv` (canonical `team_id` → `school_slug`) and `DO_NOT_ERASE/sr_school_slug_aliases.csv` (`panel_slug` → `sr_slug` for URL resolution while keeping merge keys consistent). After edits, SR refresh with `sr_refresh_do_scrape=True` only fetches **missing** `(school_slug, sr_year)` pairs in raw (resume logic), not a full re-scrape. **403** pairs are skipped via `bpm_scrape_skip_pairs.csv` until removed.

Potential placement:

- Data / linkage section (ESPN vs SR)
- Limitations (non-random missing SR advanced by institution type)
- Future work: conference- or success-based imputation of teammate quality where SR is absent

Key points:

- Unmatched SR is often institutional, not random — affects inference if draft-relevant talent concentrates in thin coverage cells.
- Crosswalk vs alias: crosswalk changes the canonical slug; alias fixes the URL when the panel slug should stay stable.
- Re-scrape is incremental by design; renaming a slug creates **new** job keys — old raw rows under the old slug do not automatically attach.

Ventile EDA vs Survival / CIF Framing (Officers Comparison)

Topic: What the inverted-U / binned draft-rate plots are and are not.

Content to consider including:

The 530 ventile figure bins **cross-sectional** `poolq_loo` (leave-one-out mean teammate `perf` within team-season) and plots mean `Y_draft` per bin vs mean `poolq_loo` — empirical draft **rate**, not time-to-event. Optional overlay is **quadratic LPM** in `poolq_loo` on the micro rows. This is **not** a cumulative incidence or survival curve; comparison to the officers workflow (CIF by bin, then bar chart of terminal values) is a **design choice** distinction worth one clear paragraph in methods. `poolq_binning` supports `quantile` (~equal n per bin; “ventiles” when `ventiles=20`) and `equal_width` on `poolq_loo`, with export slugs disambiguating runs.

Potential placement:

- Methods (nonparametric binning vs survival)
- Discussion when contrasting Scout to Army analyses
- Limitations (no time path to draft within season-year structure)

Key points:

- Each row is player–season–team; `poolq_loo` is within that season’s roster context.
- Y-axis is bin-level **draft rate**, not CIF at a horizon.
- Binning mode and z-scoring `perf` within season change interpretation of `poolq_loo` units.

Performance Measures: Box vs SR and LOO Interpretation

Topic: What enters `perf` and teammate pool quality; minutes / PPM sources.

Content to consider including:

`minutes`, `points`, `ppm`, `games` come from **ESPN game-level box** aggregated to season-team totals (PPM = total points / total minutes). **SR raw** (`bpm_player_season_raw.csv`) is the **advanced** table: `MP`, `G`, `GS`, BPM family, PER, WS, etc. — **no SR-derived PPM** in this scrape. Panel merge exposes `mp_sr` separately from box `minutes`. `perf_metric` and optional `perf_zscore_within_season` determine what is copied into `perf` before LOO; `poolq_loo` is always defined from teammates’ `perf`, not from a career average across seasons.

Potential placement:

- Variable construction / data integration
- Robustness (re-run key specs across `perf_metric` choices; filenames already slugged)

Key points:

- Same athlete can contribute multiple rows (multiple seasons / teams); LOO is **within** team-season.
- SR coverage gaps leave `perf` missing for SR-backed metrics unless merged.

`min_minutes` **Floor — Ventile Sensitivity (2026-07-17)**

Topic: How the playing-time floor changes who enters the panel, who enters LOO, and whether the inverted-U survives.

Content to consider including:

With `use_prebuilt_panel_csv=False`, `min_minutes` is applied in `panel_rebuild.build_from_box` before LOO — so it defines **both** the plotted sample and the **teammate pool** used to compute `poolq_loo`. Sensitivities on the same spec (quantile bins, winsor `(0.01, 0.99)`, ppm z-scored, `restrict_teams_by_draftees=False`, 2011–2021):

<code>min_minutes</code>	Story (Charles, Jul 2026)
0	Adds ~21k mostly non-drafted deep-bench rows (82,893 vs 62k at 20); ppm and LOO noisy; ventile curve wobbly , elite dip attenuated — dilution, not falsification. Ever-drafted count barely moves (~1,136).
10	+~10.5k player-seasons vs 20; still only +2 ever-drafted rows. With 20 quantile ventiles : elite downturn spread across bins 18→19→20 (multi-bar redundancy); noisier middle ventiles; bin 20 ~1.1% draft rate.
20	Hero candidate (provisional) : rotation-level peers in LOO. With 20 quantile ventiles : rise → plateau bins 15–18 (~2.4–2.5%) → single sharp cliff at bin 20 (~0.9%); cleaner mechanism read than 10.
50 → 100	Sample collapses toward star/rotation heavy minutes only ; inverted-U largely washes out — draft rate vs poolq looks flatter; “roster pressure” variation disappears because everyone left is a high-minute, high-talent type and teammate pools homogenize among elites.

Charles observation — 10 vs 20 at 20 ventiles (2026-07-17): Same `poolq_binning='quantile'`, `ventiles=20`, `restrict_teams_by_draftees=False`. Lowering `min_minutes` from 20 → 10 does not add meaningful draft outcomes; it adds **10–19 minute** roster filler who distort LOO and reshuffle ventile ranks. **Bin 20 draft rate ~1.1% in both specs** — the **elite ventile dip is robust**; what changes is *presentation*: **10 min** = gradual 18–20 step-down (good for “redundant downturn” visuals); **20 min** = high plateau then **one** top-ventile cliff (good for “congestion in the highest peer tier” prose). Prefer **20** for Alex hero + paper main text; **10** (and **0**) in robustness appendix.

Model read (generative): If congestion bites when **comparing across peer environments** (mid vs elite LOO pools), stripping the roster down to **only stars** removes the **cross-sectional contrast** the mechanism needs — so **attenuation at high** `min_minutes` **is consistent with the model**, not a contradiction. Low floor adds noise; very high floor removes heterogeneity.

Potential placement:

- Methods (sample definition: rotation vs full roster vs star-only)
- Robustness appendix (`min_minutes` ∈ {0, 10, 20, 50, 100})

- Discussion linking empirical sensitivity to sim “who counts in the pool”

Key points:

- `min_minutes` is not a neutral filter — it defines the estimand and LOO algebra jointly when rebuilding from box.
- **Hero lock (provisional):** see **Hero plot candidates — three-way comparison** below — `min_minutes=20` + 16 quantile + winsor `(0.01, 0.99)` + `use_prebuilt_panel_csv=False`.
- **High floor = star-only sensitivity:** document as “congestion signal requires roster depth variation.”
- **One-liner for Alex:** “Draft rate rises with LOO teammate quality, then drops in the **highest ventile**; dip stable at 10 and 20 minutes, washes out when we star-select or include full deep bench.”

Status: Export PNG/CSV triplet for `{0, 10, 20, 50, 100}` × `{16, 20}` quantile when archiving robustness bundle; copy locked hero to `alex_side-by-side_v0/` when final.

Hero Plot Candidates — Three-Way Comparison (2026-07-17)

Topic: Charles compared three ventile EDA specs on the **same rebuilt panel** before locking the Alex side-by-side hero. Shared base for all three:

- `use_prebuilt_panel_csv=False`, `min_minutes=20`, `poolq_winsor_quantiles=(0.01, 0.99)`
- `perf_metric=ppm`, `perf_zscore_within_season=True`, `restrict_teams_by_draftees=False`
- Seasons 2011–2021 → 62,180 player-seasons, 1,134 with `Y_draft=1`
- Only fork: `poolq_binning` × `ventiles`

Content to consider including:

Candidate	Spec	Shape (Charles, Jul 2026)	Per-bin n	Verdict
1	8 equal-width	Low flat left → rise bins 4–6 (peak ~2.3–2.4%) → one drop (bin 7 ~1.1%) → bin 8 ~0	Extremely uneven — e.g. bin 3 22k, bin 7 1,145	Coarse sensitivity only; elite end = one noisy bar + empty bar
2	16 equal-width	Long climb → peak ~bin 10 (~2.5%) → several declining bars (11→16)	Wild spread — e.g. bin 8 9,712, bin 15 301, bin 16 844	Matches “≥2 bars on downturn” visually, but tail bins unreliable (see below)
3	16 quantile	Steady rise bins 2–12 → plateau bins 12–15 (~2.3–2.6%) → sharp drop only at bin 16 (~1.2%)	~3,886–3,887 every bin	Recommended hero — clearest inverted-U story, equal precision per bar

Charles observation — why 16 equal-width sends the last 2 bins to 0:

Switching from `quantile` to `equal_width` with `ventiles=16` (same `min_minutes=20`, same winsor) does **not** just “zoom in” on the elite tail — it **re-partitions the x-axis** on `poolq_loo` units via `pd.cut` (equal intervals from min to max winsorized LOO). After z-scoring within season, most player-seasons still pile up in the **middle** of the LOO distribution; equal-width bins therefore get **enormous n in the center** and **sparse n at both tails**.

Exported CSV (`...poolqeqwidth...2026-07-17.csv`) for candidate 2:

Bin (1-indexed)	n	Mean draft rate	Mean <code>poolq_loo</code>
14	433	~0.23%	~0.78
15	301	0.0%	~0.91
16	844	0.0%	~1.09

So the **last two equal-width bins are not empty of rows** (301 + 844 = **1,145** player-seasons in bins 15–16) — but they contain **zero drafted players** in this run. That produces a **literal cliff to 0** on the PNG, which reads like “the curve broke” rather than “congestion in the top ventile.”

Why quantile bin 16 does not do this: `qcut` /rank quantile forces ~3,886 rows per bin, including drafted rotation players, into the top **6.25%** of LOO ranks. Bin 16 still shows a dip (~1.2%, ~45 draft events expected-scale) — a **statistically stable** elite-tier downturn. Equal-width bins 15–16 instead isolate a **tiny tail slice in LOO space** where draft picks are **structurally rare** (~1,134 drafts total, concentrated below the extreme LOO cap).

Interpretation (not over-claiming):

- The **multi-bar downturn** on 16 equal-width is **partly real composition** (declining draft rate as mean LOO rises through bins 11–14) and **partly tail sparsity** — bins **15–16** hit **integer zero** because too few drafted players land in those narrow high-LOO intervals.
- This is **not** the same falsification as `min_minutes=50-100` (star-only washout); it is a **binning artifact** that equal-width invites when `poolq_loo` is bell-shaped and winsor-clipped.
- Candidate 2 remains useful as **appendix / sensitivity** if tail bars are **labeled with n** (see deferred equal-width bar-width idea below) — do **not** treat bins 15–16 at exactly 0% as standalone proof of congestion.

Recommendation (Jul 2026 Alex prep):

Priority	Pick	Why
Hero (empirical + sim side-by-side)	Candidate 3 — 16 quantile	Rise + elite dip, equal n per bar, defensible “top ventile” language, easiest sim match
Appendix	Candidate 2 — 16 equal-width	Shows multi-bar tail decline; must annotate thin-n tail
Coarse sensitivity	Candidate 1 — 8 equal-width	Too lumpy for hero

Potential placement:

- Methods (quantile vs equal-width binning choice)
- Alex slide caption + `alex_side_by_side_v0/` canonical export
- Robustness appendix (equal-width sensitivity with per-bin `n`)

Key points:

- `poolq_binning` is a **presentation / estimand choice** for EDA — not interchangeable with quantile ventiles when tails matter.
- **Equal-width + many bins + winsorized LOO** → **expect zeros or near-zeros in top bins** even when the quantile hero shows a clean ~1% top-ventile dip.
- **Sim must match hero binning rule** (16 quantile, same winsor, same `min_minutes`) — not just “poolq_loo on both axes.”

Status: Hero triplet + side-by-side bundle in `alex_side_by_side_v0/`; see `Hero_Model_Three_Layers_Memo.md` and `sports/scripts/hero_model_reset_bundle.py`.

High School Performance, Teammate Pools, and Binning (Advisor Discussion)

Topic: Enrich the panel with HS data; bin on pre-college ability while preserving leave-one-out **team** structure.

Content to consider including:

Discussion with advisor: add **high school** (or recruiting) data so players are **binned** along a dimension defined by **HS performance metrics** (e.g. composite rank, scouting grades, stats), rather than binning on `poolq_loo` built from **collegiate seasonxteam** `perf` as in the current 530 ventile EDA. The conceptual **team pool** should **remain**: still compute **leave-one-out teammate pool quality** within `(team_id, season)`, but drive `perf` (or a parallel HS column) from **high-school-level measures** so “how good are my teammates *as projected from HS?*” replaces or complements “how good are my teammates *this college season on this stat?*” Implementation requires a **merge** to `athlete_id` (or a stable recruiting ID), decisions on **missing HS** coverage, and whether HS metrics are **z-scored within cohort** (class year, position) before LOO. The ventile / LPM machinery can stay the same mechanically once `poolq_loo` is defined from the chosen HS-based `perf`.

Potential placement:

- Methods (covariate timing: all pre-college information for the binning axis)
- Data section (HS / recruiting sources, linkage rate)
- Discussion (interpretation: sorting on ex-ante talent vs realized college performance)
- Limitations (selection into observable HS data; international / JUCO paths)

Key points:

- Separates **who you are surrounded by in HS terms** from **college box/SR realization** for stratification.
 - LOO algebra unchanged: still exclude self within team-season; only the input to `perf` changes.
 - Creates a clear story for “peer pool at arrival” vs current “peer pool on realized college metric.”
 - Not yet in `sports_pipeline` — needs schema, merge QA, and sensitivity to missing HS.
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Restricting to Teams With at Least One Drafted Player

Topic: Drop rows for teams with **no** drafted player under a clear rule (program vs season).

Content to consider including:

Implemented in code (`panel_build.filter_panel`): `restrict_teams_by_draftees` (default **True**) and `draftee_restriction`: `all_time` — keep only `team_id` with ≥ 1 row with `Y_draft==1` anywhere in the filtered sample before this step; `season` — keep only `(team_id, season)` where ≥ 1 roster member has `Y_draft==1` that season. Applied after `dropna(poolq_loo, Y_draft)` and `min_minutes`, so ventiles, LPM, and integrity `use` all align. Export slugs gain `_tdalltime` or `_tdseason` when the restriction is on (see `perf_metric.export_plot_slug`). Motivation: focus draft-rate vs pool-quality on environments where draft outcomes are not structurally zero on the roster. Trade-offs: smaller sample, selection (e.g. low-majors), changed interpretation — report **counts dropped** and run **robustness** with `restrict_teams_by_draftees=False`.

Potential placement:

- Sample definition / inclusion rules
- Robustness appendix (full sample vs draft-exposed-teams-only)
- Limitations (generalization beyond teams that ever produce a draft pick)

Key points:

- Sharpens comparison sets where draft outcome is empirically possible on the roster.
- Risks truncating heterogeneity — justify and show robustness.
- `all_time` vs `season` answers “program ever had a draftee in window” vs “this season’s roster had a draftee.”

Charles observation (2026-07-17): Restricting to teams who have **ever** had a draftee (`restrict_teams_by_draftees=True`, `draftee_restriction="all_time"`) produces **startlingly different** ventile curves vs the full sample (`restrict_teams_by_draftees=False`). The shift is **even larger** with `draftee_restriction="season"` — keep only `(team_id, season)` rosters that actually had a draft pick that year. Treat this as a **major sample-definition fork**, not a minor robustness tweak: the inverted-U shape, tail bins, and draft rates can move a lot. **Alex hero plot** currently uses `restrict_teams_by_draftees=False`; any draftee-restricted run is a **different estimand** and must be labeled explicitly in provenance and side-by-side exports.

Status: Sensitivity branch — compare full sample vs `all_time` vs `season` in appendix; do not silently swap for the canonical empirical curve without noting it.

Sorting Noise Robustness Thought

Topic: Later robustness version for noisy assortative sorting in `537_Sports_Simulation.ipynb`.

Content to consider including:

For the first noisy-sorting implementation, use raw Gaussian sorting noise:

`sorting_signal_i = A_i + epsilon_i`, where `epsilon_i ~ Normal(0, SORTING_NOISE_SD)`.

This keeps the first version easy to explain: true ability `A_i` is unchanged, but the pool assignment process observes a noisy version of ability. When `SORTING_NOISE_SD = 0`, sorting is perfectly assortative; as it increases, pool assignment becomes less perfectly ordered.

Later, revisit a relative-noise version:

`epsilon_sd = SORTING_NOISE_FRACTION * sd(A_i)` .

That version may be more portable across different ability distributions because the noise scale adjusts to the observed spread of ability. Hold this for a robustness/sensitivity branch after the raw-noise version is clear.

Potential placement:

- Simulation appendix or robustness subsection
- Notes around noisy sorting in `537_Sports_Simulation.ipynb`

Key points:

- Noise is **only for sorting into pools**, not for true ability or promotion weights.
- Raw `SORTING_NOISE_SD` is easier to teach first.
- Relative `SORTING_NOISE_FRACTION * sd(A_i)` is the later robustness idea to remember.

Equal-Width Ventile Bars: Encode Bin n Visually (2026-07-17)

Topic: When `poolq_binning="equal_width"` , bar counts per bin are unequal (unlike quantile/`qcut` ventiles). The draft-rate height alone can misread precision in thin tail bins.

Content to consider including:

For `ventile_eda_plot_style="bins_bars_520"` (530 CELL 4 / `panel_build.ventile_plot`), optionally encode `n` from the binned table (`ventile_table` → column `n`) in the bar geometry or color — e.g. **narrower or lighter bars** when a bin has fewer player-seasons, **full width / saturated color** when `n` is large. Goal: reader sees at a glance that equal **x-width** ≠ equal **sample size** (especially bins 1–2 and 14–16 on the current 16-bin ppm spec).

Potential placement:

- 530 ventile PNG polish (methods figure or appendix)
- Same pattern for generative side-by-side plots once sim bins are exported with counts

Key points:

- **Deferred** — do not block Alex side-by-side / `poolq_loo` sim work (Jul 2026).
- Quantile binning already approximates equal `n` ; this idea matters mainly for **equal_width** runs.
- CSV already has per-bin `n` ; change is plotting-only in `ventile_plot` .

Status: Idea logged; not implemented.

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- CSV already has per-bin `n`; change is plotting-only in `ventile_plot`.

Status: Idea logged; not implemented.

Small Things to Check Later

- Confirm whether `Y_draft` should be “ever drafted” vs draft **in or after** season k for causal timing stories.
- Document any manual edits to `sr_school_slug_crosswalk.csv` in a one-line changelog row or git commit message when possible.
- `export_panel_after_run` during multi-metric sweeps overwrites one path — document if a dated archive is needed.
- Implement and document **HS → athlete** merge keys; add `poolq_loo_hs` (or swap `perf` source) when data land.
- Log **N excluded** by `restrict_teams_by_draftees` in a one-off QA cell or export (optional enhancement).